

SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
BTECH DEGREE II SEMESTER FIRST INTERNAL EXAMINATION, APRIL 2024
SUBJECT: 23-200-0205A BASIC ELECTRONICS ENGINEERING (2023 SCHEME)

Duration: 2 HOURS

ME-A / SFE

Maximum Marks: 30

Course Outcomes:

On successful completion of the course, the students will be able to:

- CO 1: Illustrate the concept of diode in rectifiers, filter circuits and wave shaping.
 CO 2: Interpret the functioning of oscillators and operational amplifiers.
 CO 3: Explain the principles of embedded systems and sensors.
 CO 4: Summarize the functioning of a communication system and different modulation technologies

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply,
 L4 – Analyze, L5 – Evaluate, L6 – Create

PART - A

(Answer all questions)

(4 x 2 = 10)

Q. No.	Question	Marks	BL	CO	PO
I. a)	What is the role of reservoir capacitor in a half wave rectifier?	2	L2	1	1
b)	An amplifier produces an open-loop gain of 180. Determine the amount of feedback required if it is to be operated with a precise voltage gain of 50	2	L3	1	1
c)	A mains transformer having a turns ratio of 44:1 is connected to a 220 V r.m.s. mains supply. Explain the concept of turns ratio in the given context?	2	L3	1	1
d)	Explain different types of amplifiers.	2	L1	1	1
e)	What do you mean by operational amplifier? Explain its symbol in detail.	2	L1	2	1

PART - B

(2 x 10 = 20)

Q. No.	Question	Marks	BL	CO	PO
II a)	Describe any one full wave rectifier. Also explain the methods used to reduce the ripples at the output of full wave rectifier with neat diagrams.	6	L1	1	1
b)	Explain the need of Voltage regulators in power supply systems with neat circuit diagram?	4	L2	1	1
OR					
III a)	Explain the terms bandwidth and frequency response of an amplifier?	5	L2	1	1
b)	Elaborate the following parameters of amplifiers				
	• Gain	5	L1	1	1
	• Negative feedback				
	• Phase shift				

- IV** a) Describe the three main configurations of operational amplifier with appropriate diagrams
- b) Explain the VI characteristics of a PN junction diode with necessary diagrams.

6 L1 2

4 L1 1

OR

- V** a) List all the characteristics associated with an operational amplifier and compare the ideal and practical values of the characteristics in a table.
- b) An operational amplifier with negative feedback applied produces an output of 1.5 V when an input of 7.5 mV is present. Determine the closed-loop voltage gain.

8 L2 2 1

2 L3 2 1

L1=50% L2=38% L3=12%